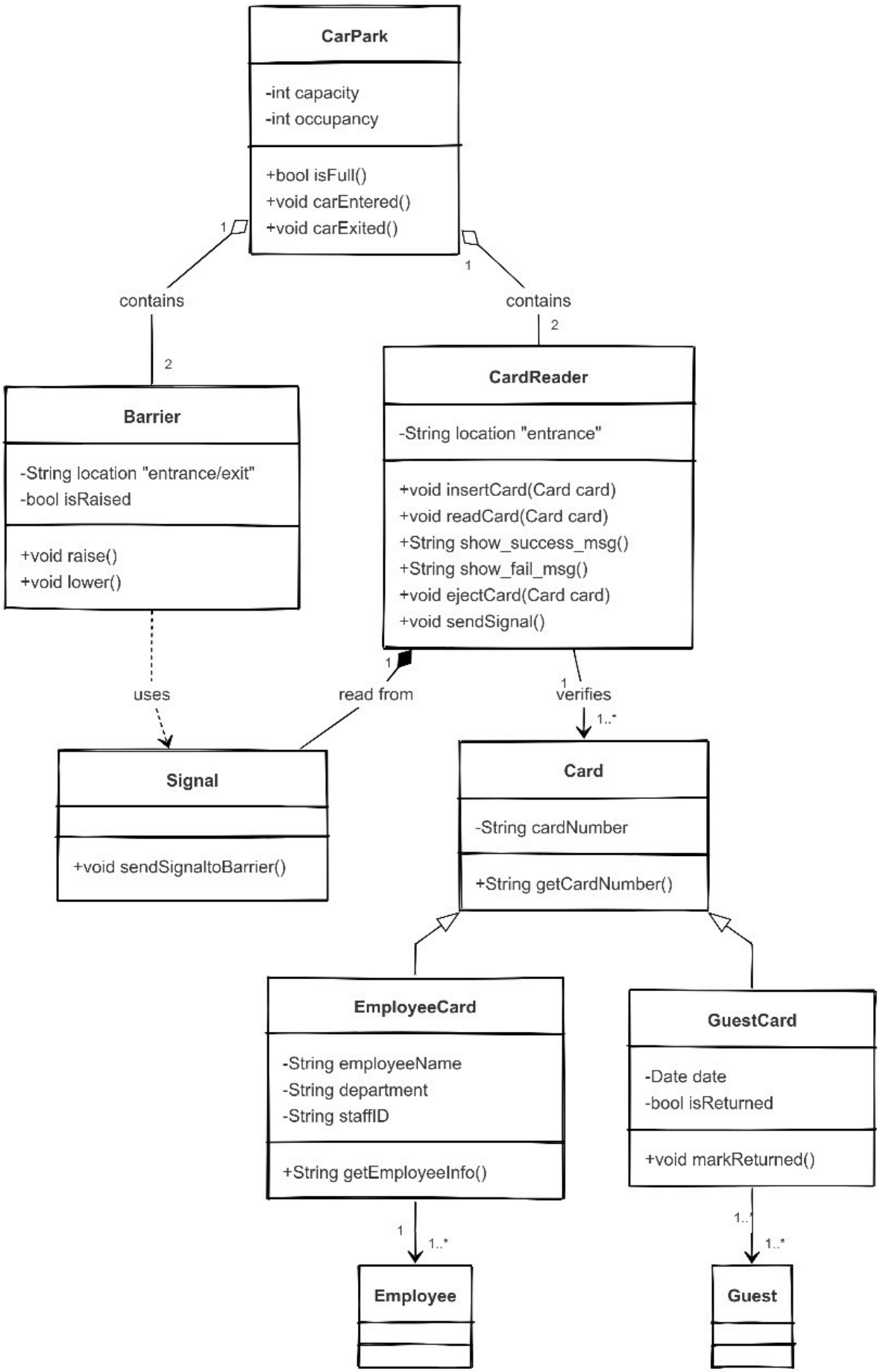
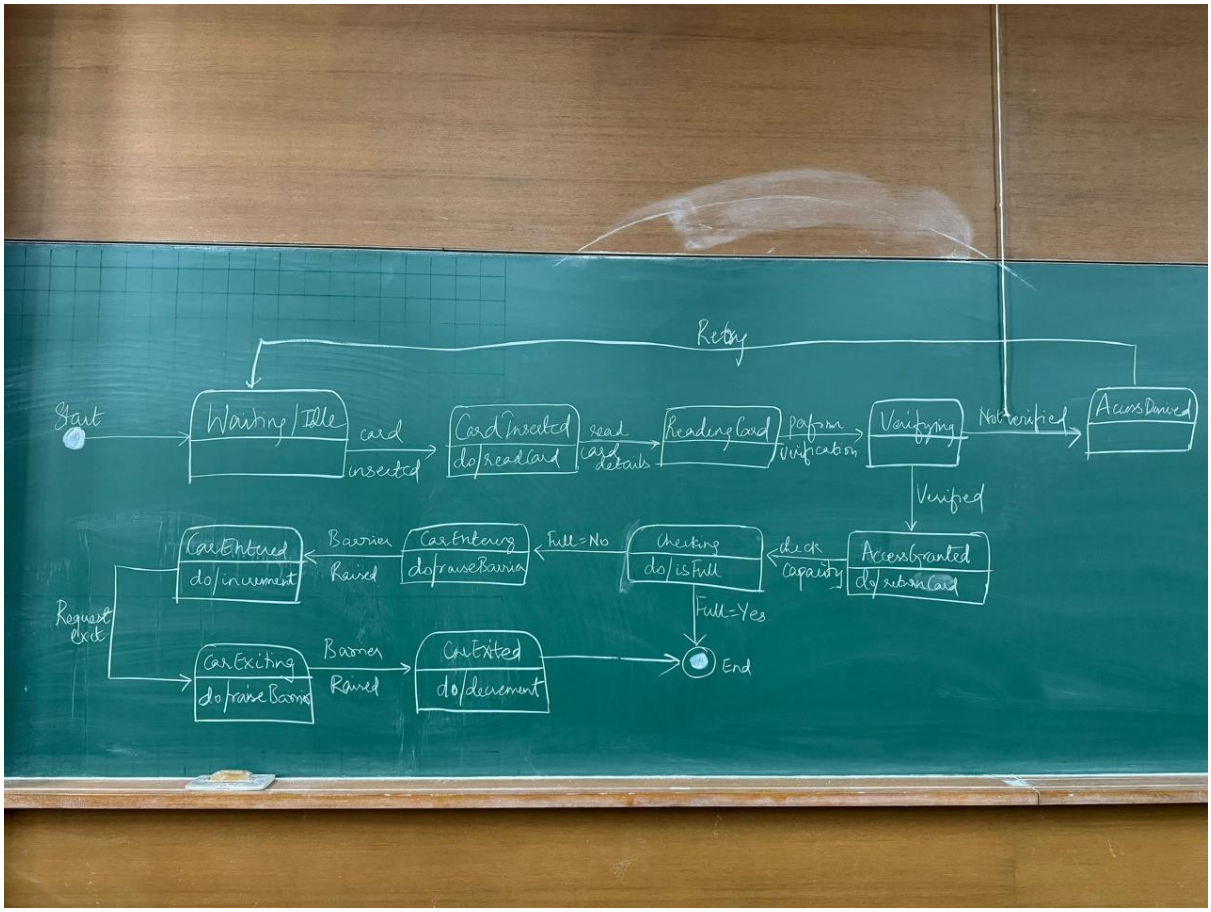
Example

- Saturn Int. management wants to improve their security measures, both for their building and on site. They would like to prevent people who are not part of the company to use their car park.
- Saturn Int. has decided to issue identity cards to all employees. Each card records the name, department and number of a company staff, and give them access to the company car park. Employees are asked to wear the cards while on the site.
- There is a barrier and a card reader placed at the entrance to the car park. When a driver drives his car into the car park, he/she inserts his or her identity card into the card reader. The card reader then verify the card number to see if it is known to the system. If the number is recognized, the reader sends a signal to trigger the barrier to rise. The driver can then drive his/her car into the car park.
- There is another barrier at the exit of the car park, which is automatically raised when a car wishes to leave the car park.
- A sign at the entrance display “Full” when there are no spaces in the car park. It is only switched off when a car leaves.
- There is another type of card for guests, which also permits access to the car park. The card records a number and the current date. Such cards may be sent out in advance, or collected from reception. All guest cards must be returned to reception when the visitor leaves Saturn Int.

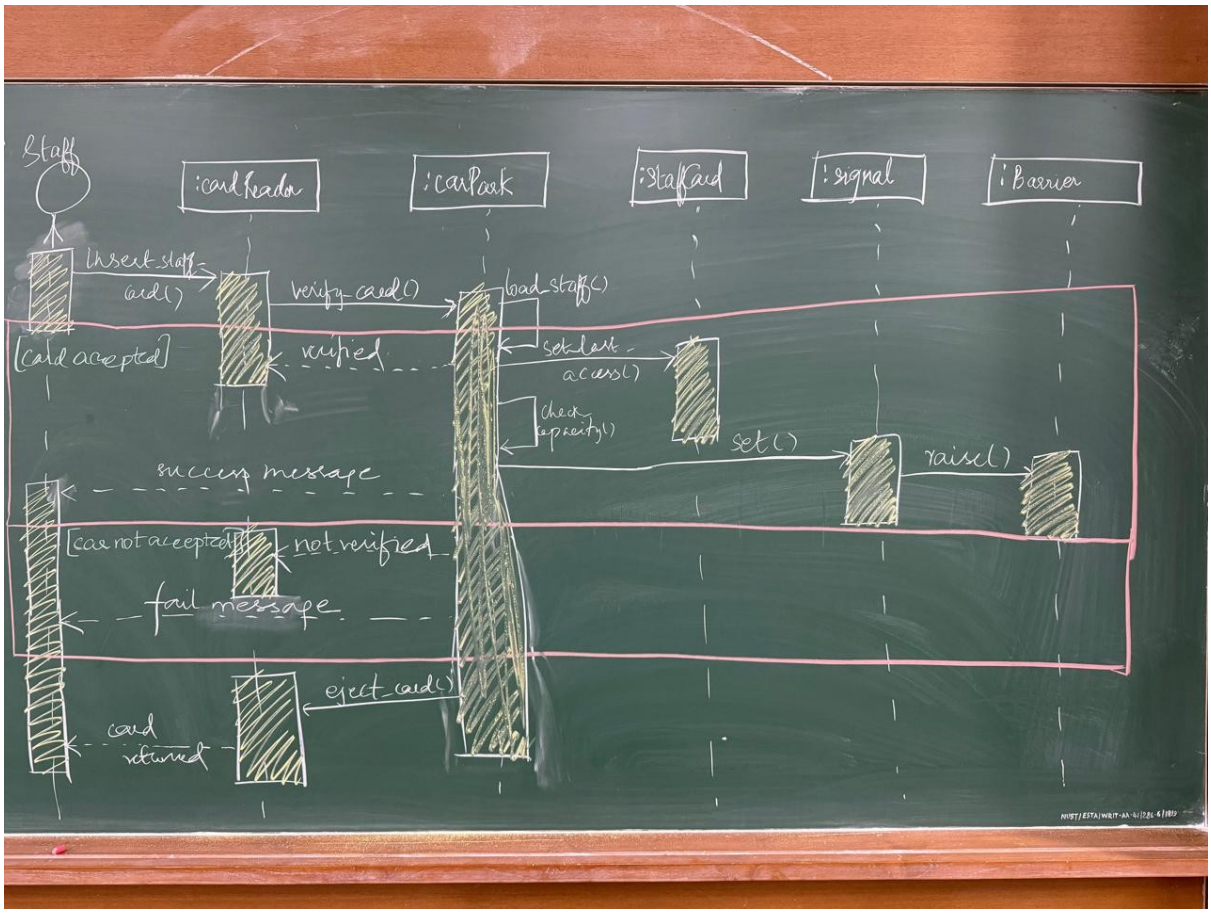
Class Diagram:


CarPark
-int capacity
-int occupancy
+bool isFull()
+void carEntered()
+void carExited()
1
contains
2
1
contains
2
Barrier
-String location "entrance/exit"
-bool isRaised
+void raise()
+void lower()
CardReader
-String location "entrance"
+void insertCard(Card card)
+void readCard(Card card)
+String show_success_msg()
+String show_fail_msg()
+void ejectCard(Card card)
+void sendSignal()
uses
1
read from
1
verifies
1..*
Signal
+void sendSignaltoBarrier()
Card
-String cardNumber
+String getCardNumber()
EmployeeCard
-String employeeName
-String department
-String staffID
+String getEmployeeInfo()
GuestCard
-Date date
-bool isReturned
+void markReturned()
1
1..*
1..*
1..*
Employee
Guest

State Diagram:



Sequence Diagram:


Staff
:cardReader
:carPark
:StaffCard
:signal
:Barrier
insert_staff_card()
verify_card()
load_staff()
[card accepted]
verified
set_last_access()
check_property()
set()
raise()
success message
[car not accepted]
not_verified
fail message
eject_card()
card returned